

Register Transfer & Micro-Operations

Q. What are Micro-Operations? Discuss the four main types: Register Transfer, Arithmetic, Logic, and Shift Micro-Operations with examples.

📌 Definition to Remember

A **Micro-Operation** is the most basic, fundamental, atomic operation performed on data stored in CPU registers during one clock pulse. Every high-level machine instruction is broken down into a sequence of these elementary micro-operations.

1. Types of Micro-Operations

There are four primary categories of micro-operations based on their function.

A. Register Transfer Micro-Operations

Transfers data from one register to another without modifying the actual data.

* **Notation:** `<-` indicates transfer.

* **Example:** `R2 <- R1` (Copies data from R1 into R2. R1 is unchanged).

* **Control:** Usually triggered by a control condition: `If (P=1) then R2 <- R1`.

B. Arithmetic Micro-Operations

Performs basic math on numeric data in registers.

* **Operations:** Add, Subtract, Increment, Decrement, 2's Complement (Negation).

* **Examples:**

* `R3 <- R1 + R2` (Addition)

* `R1 <- R1 + 1` (Increment)

* `R2 <- R1' + 1` (2's Complement/Negate R1, store in R2).

C. Logic Micro-Operations

Performs bit-by-bit manipulation on non-numeric binary data.

* **Operations:** AND, OR, XOR, NOT.

* **Uses:** Masking bits, clearing registers, inserting bits.

* **Examples:**

* `R1 <- R1 ^ R2` (Bitwise XOR).

* `R1 <- R1 & R2` (Bitwise AND).

* `R1 <- 0` (Can be done by XORing a register with itself: `R1 <- R1 ^ R1`).

D. Shift Micro-Operations

Moves the binary bits of a register left or right. Used for data alignment and fast multiplication/division.

* **Logical Shift:** Shifts bits, filling the newly empty space with `0`. (`R1 <- shl R1`).

* **Circular Shift (Rotate):** Bits shifted out of one end are fed directly back into the other end.

* **Arithmetic Shift:** Shifts right but preserves the Most Significant Bit (MSB/Sign bit), essential for signed negative numbers.

★ Must-Write Points (for 10 marks)

1. **Micro-Operation:** The smallest atomic operation executed in one clock pulse.
2. Four types: **Register Transfer, Arithmetic, Logic, Shift.**
3. **Register Transfer:** Moves data without changing it. Example: `R2 <- R1`.
4. **Arithmetic:** Basic math (Add, Sub, Inc). Example: `R3 <- R1 + R2`.
5. **Logic:** Bitwise operations (AND, OR, XOR) used for masking. Example: `R1 <- R1 ^ R2`.
6. **Shift:** Moves bits left/right for alignment or fast math.
7. Shift types: Logical (fills with 0), Circular (rotates bits), Arithmetic (preserves sign bit).

⚡ Quick Recall

Micro-Ops (1 clock pulse) → 4 Types: Transfer (`R2<-R1`) → Arithmetic (`R1+R2, +1`) → Logic (AND/XOR, bit masking) → Shift (Logical, Circular, Arithmetic for signed math)